

MEASURING DATA SHEET (2)				Customer																					
Measuring date						Measuring person																			
Measuring position																									
DUST CONCENTRATION IN FLUE GAS								Calculated by	Checked by	PC Input	Collated by														
Circular filter paper				Cylindrical filter paper																					
☐ Form 1 ☐ Form 2		☐ Form 1 ☐ Form 2																							
☐ QMA ☐ PF-020		☐ Whatman ☐ 89		Atmospheric pressure				P_a		kPa															
		☐ 89S		Oxygen standard concentration				O_n		%															
Measuring points	Measuring time	Number of points	Diameter of nozzle d mm	Filter paper No.	Suction duration t min	Gas meter			Suction volume at standard condition V_s m ³	Mass of collected dust m g	Dust concentration		Oxygen concentration O_s %												
						Suction volume V L	Temp. θ_m °C	Pressure P_m kPa			Measured value C_s g/m ³	Conversion value C g/m ³													
										Average															
Equipment No.	Gas meter		DVM — 01 —				Suction pump		DVM — —																
	Orsat apparatus		DVM — 045 —																						
								Calculated by	Checked by	PC Input	Collated by														
NITROGEN OXIDE CONCENTRATION IN FLUE GAS																									
Measuring point						Oxygen standard concentration		O_n		%															
Measuring time	NO _x flask								Suction volume at standard condition V_s mL	Ion concentration determined from the working curve		Nitrogen oxide concentration		Oxygen concentration O_s %											
	No.	Before sampling			After sampling			Volume $V-20$ mL		NO ₃ ⁻ a mg/L	NO ₂ ⁻ b mg/L	Measured value C_s ppm	Conversion value C ppm												
		Absolute pressure P_i kPa	Temp. t_i °C	Saturated vapor pressure P_{ni} kPa	Absolute pressure P_f kPa	Temp. t_f °C	Saturated vapor pressure P_{nf} kPa																		
										Average															
Flask (After collecting) <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th>No.</th> <th>Atmospheric pressure kPa</th> <th>Residual pressure kPa</th> </tr> </thead> <tbody> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> </tbody> </table>														No.	Atmospheric pressure kPa	Residual pressure kPa									
No.	Atmospheric pressure kPa	Residual pressure kPa																							
Equipment No.	Manometer		DVM — —				Mighty pump		DVM— 029 —																
	Orsat apparatus		DVM — 045 —																						
Retention period	5 years	Storage department	Technical room	TEC-G-QT-001-003	Revised (Enacted) date	1-Apr-2015																			

MEASURING DATA SHEET (3)				Customer			
Measuring date				Measuring person			
Measuring position							
Calculated by		Checked by	PC Input	Collated by			
SULFUR OXIDE CONCENTRATION IN FLUE GAS				Atmospheric pressure		P_a	kPa
				Oxygen standard concentration		O_n	%
Measuring point	Times	Measuring time	Gas meter			Suction volume at standard condition	Oxygen concentration
			Suction volume	Temperature	Pressure		
			V	θ_m	P_m		
			L	°C	kPa	L	%
	①						
	②						
	③						
Equipment No.	Gas meter	DVM — 01 —			Suction pump	DVM — —	
	Barometer	DVM — 016 —					

1. Analysis method: Ion chromatography method

Times	SO ₄ ²⁻ concentration determined from the working curve	Prepared volume	Sulfur oxide concentration		Oxygen concentration
			Measured value	Conversion value	
	a		C_s	C	
	mg/L	mL	ppm	ppm	%
①					
②					
③					
		Average			

2. Analysis method: Precipitation titration method

5mmol/L Ba(CH₃COO)₂ solution

f

Times	Titrated value			Analysis sample solution		Sulfur oxide concentration		Oxygen concentration
	Sample test	Blank test	Difference	Prepared volume	Aliquot	Measured value	Conversion value	
	a	b	$a - b$	v_1	v_2	C_s	C	
	mL	mL	mL	mL	mL	ppm	ppm	%
①								
②								
③								
				Average				

3. Analysis method: Turbidimetric method

Working curve

K

mg/Abs

Times	Sample absorbance	SO ₄ ²⁻ concentration determined from the working curve	Analysis sample solution		Sulfur oxide concentration		Oxygen concentration
			Prepared volume	Aliquot	Measured value	Conversion value	
		a	v_1	v_2	C_s	C	
		mg	mL	mL	ppm	ppm	%
①							
②							
③							
			Average				

4. Fuel component, a method for calculation from the used amount

Type of fuel	Density of the fuel	Sulfur content of fuel	Amount of used fuel	Amount of emission
	r	C_s	W	q
	kg/L	%	kL/h	m ³ /h

Calculation formula

$$q = \frac{22.41}{32} \times \frac{C_s}{100} \times W \times \rho \times 1000$$

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MEASURING DATA SHEET (4)				Customer						
Measuring date				Measuring person						
Measuring position										
		Calculated by	Checked by	PC Input	Collated by					
						Atmospheric pressure	P_a kPa			
						Oxygen standard concentration	O_n %			
HYDROGEN CHLORIDE CONCENTRATION IN FLUE GAS										
Measuring point	Times	Measuring time	Gas meter			Suction volume at standard condition	Remaining oxygen concentration			
			Suction volume	Temperature	Pressure					
			V L	θ_m °C	P_m kPa					
	①									
	②									
	③									
Equipment No.	Gas meter	DVM — 01 —			Suction pump	DVM — —				
	Barometer	DVM — 016 —								
1. Analysis method: Ion chromatography method										
Times	Cl ⁻ concentration determined from the working curve		Prepared volume	Hydrogen chloride concentration				Oxygen concentration		
				Measured value		Conversion value				
	a mg/L			C_s mg/m ³	ppm	C mg/m ³	ppm		O_s %	
①										
②										
③										
			Average							
2. Analysis method: Silver nitrate titration method										
			0.1mol/L NH ₄ SCN solution				f			
Times	Titrated value			Analysis sample solution		Hydrogen chloride concentration				Oxygen concentration
	Sample test	Blank test	Difference	Prepared volume	Aliquot	Measured value		Conversion value		
	a mL	b mL	$b - a$ mL	v_1 mL	v_2 mL	C_s mg/m ³	ppm	C mg/m ³	ppm	
	a mL	b mL	$b - a$ mL	v_1 mL	v_2 mL	C_s mg/m ³	ppm	C mg/m ³	ppm	
①										
②										
③										
				Average						
3. Analysis method: Mercury (II) thiocyanate spectrophotometry										
			Working curve		K	mg/Abs				
Times	Sample absorbance	SO ₄ ²⁻ concentration determined from the working curve	Analysis sample solution		Hydrogen chloride concentration				Oxygen concentration	
			Prepared volume	Aliquot	Measured value		Conversion value			
			v_1 mL	v_2 mL	C_s mg/m ³	ppm	C mg/m ³	ppm		
			a mg		C_s mg/m ³	ppm	C mg/m ³	ppm		
①										
②										
③										
			Average							
Retention period	5 years	Storage department	Technical room	TEC-G-QT-001-003	Revised (Enacted) date	1-Apr-2015				

MEASURING DATA SHEET (5)		Customer	
Measuring date		Measuring person	
Measuring position			

Calculated by	Checked by	PC Input	Collated by

SULFUR OXIDE ・ HYDROGEN CHLORIDE CONCENTRATION IN FLUE GAS

(Analysis method: Ion chromatography method)

Atmospheric pressure	P_a	kPa	
Oxygen standard concentration	O_n	%	

Measuring point	Times	Measuring time	Gas meter			Suction volume at standard condition	Oxygen concentration
			Suction volume	Temperature	Pressure		
			V L	θ_m °C	P_m kPa		
	①						
	②						
	③						
Equipment No.	Gas meter	DVM — 01 —				Suction pump	DVM — —
	Barometer	DVM — 016 —					

1. Sulfur oxide concentration

Times	SO ₄ ²⁻ concentration determined from the working curve	Prepared volume	Sulfur oxide concentration		Oxygen concentration
			Measured value	Conversion value	
			C_s ppm	C ppm	
①	a mg/L	v mL			O_s %
②					
③					
		Average			

2. Hydrogen chloride concentration

Times	Cl ⁻ concentration determined from the working curve	Prepared volume	Hydrogen chloride concentration				Oxygen concentration
			Measured value		Conversion value		
			C_s mg/m ³	ppm	C mg/m ³	ppm	
①	a mg/L	v mL					O_s %
②							
③							
		Average					

Retention period	5 years	Storage department	Technical room	TEC-G-QT-001-003	Revised (Enacted) date	1-Apr-2015
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MEASURING DATA SHEET (6)			Customer	
Measuring date		Measuring person		
Measuring position				

Calculated by	Checked by	PC Input	Collated by

SULFUR OXIDE · NITROGEN OXIDE · HYDROGEN CHLORIDE CONCENTRATION IN FLUE GAS

(Analysis method: Ion chromatography method)

										Oxygen standard concentration	O_n	%	
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Measuring point	Measuring time	NO _x flask										Suction volume at standard condition
		No.	Before sampling			Atmospheric pressure kPa	Residual pressure kPa	After sampling			Volume V-20 mL	
			Absolute pressure	Temp.	Saturated vapor pressure			Absolute pressure	Temp.	Saturated vapor pressure		
			P_i	t_i	P_{ni}			P_f	t_f	P_{nf}		
			kPa	°C	kPa			kPa	°C	kPa		

Equipment No.	Manometer	DVM — —	Mighty pump	DVM — 029 —
	Orsat apparatus	DVM — 016 —		

1. Sulfur oxide concentration

Number of times	SO ₄ ²⁻ concentration determined from the working curve	Prepared volume v mL	Sulfur oxide concentration		Oxygen concentration O_s %
			Measured value C_s ppm	Conversion value C ppm	
	a mg/L				
①					
②					
③					
Average					

2. Nitrogen oxide concentration

Number of times	Ion concentration determined from the working curve		Nitrogen oxide concentration		Oxygen concentration O_s %
	NO ₃ ⁻	NO ₂ ⁻	Measured value C_s ppm	Conversion value C ppm	
	a	b			
	mg/L	mg/L			
Average					

3. Hydrogen chloride concentration

Number of times	Cl ⁻ concentration determined from the working curve	Prepared volume v mL	Hydrogen chloride concentration				Oxygen concentration O_s %
			Measured value C_s		Conversion value C		
	a		mg/m ³	ppm	mg/m ³	ppm	
	mg/L						
①							
②							
③							
Average							

Retention period	5 years	Storage department	Technical room	TEC-G-QT-001-003	Revised (Enacted) date	1-Apr-2015
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Measuring position:

Measuring condition:

Analysis method: Ion chromatography method

[illegible]

Equipment No.		
Suction equipment system No.	Gas meter	Diaphragm pump
1		
2		
3		
4		

Calculated by	Checked by	PC Input	Collated by

Measuring person:

Retention period	5 years	Storage department	Technical room	TEC-G-QT-001-003	Revised (Enacted) date	1-Apr-2015
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Measuring position:

Measuring condition:

Analysis method: Indophenol blue absorption spectrophotometry method

$$K = \text{mgNH}_4^+/\text{Abs}$$

Equipment No.		
Suction equipment system No.	Gas meter	Diaphragm pump
1		
2		
3		
4		

Measuring person:

Retention period	5 years	Storage department	Technical room	TEC-G-QT-001-003	Revised (Enacted) date	1-Apr-2015
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